

Predictive IoT Framework for Early Post-Harvest Spoilage Detection

Sub-Theme: Food, Agriculture, and Health

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The Silent Crisis



The Illusion of Safety

Current warehouse inspections rely entirely on human vision. By the time decay is visible, microscopic pathogens have already multiplied.

The Chain Reaction

A single unnoticed rotting sack rapidly cross-contaminates surrounding healthy stock, leading to massive agricultural food waste.

The Hidden Hazard

If micro-spoilage escapes the warehouse, it introduces severe, silent biological toxins into our daily food supply, posing critical health risks to consumers.



A Paradigm Shift: **Smelling** **Over Seeing**

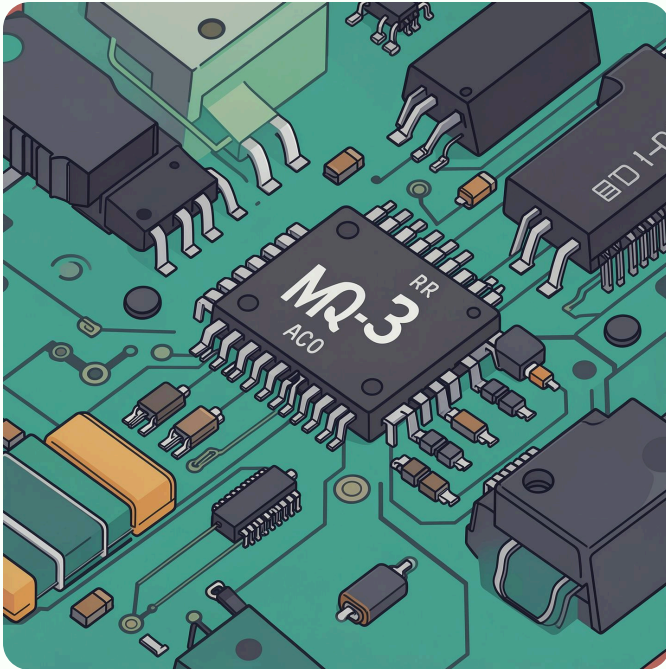
The Biochemical Reality

Rot begins chemically long before it appears physically. Decaying crops release volatile organic compounds (VOCs) and ethanol gases **28-36 hours** prior to visible fungal growth.

The Shift

Defense must move from sight to off-gassing detection. **If you can see the rot, it is already too late.** If you can smell it electronically, you can prevent it.

Scalable Architecture & Economic Feasibility



Core Components

Powered by an Arduino microcontroller and a high-sensitivity MQ-3 VOC sensor – a proven, low-cost hardware pairing.



Economic Feasibility

Ultra-low-cost architecture ensures extreme affordability for local farmers and mandi operators across India.



Calculated Coverage

1 Sensor Node effectively monitors **~150 sq. ft.** of dense rack space – approximately 80 - 100 standard grain sacks.

The Predictive Framework: System Overview



From physical grain to smartphone alert – the system intercepts spoilage before food is lost.

Phase 1: SENSE

The MQ-3 node continuously monitors the localized microclimate for VOC spikes in real time.

Phase 2: ANALYZE

Data streams to a Python-based Random Forest ML model, which analyzes historical trends to filter out false alarms.

Phase 3: PREDICT & PREVENT

The AI calculates a live **Spoilage Risk %**. If critical, it triggers automated alerts, intercepting the spoilage chain before the food is lost.